



Iron

Fact Sheet for Health Professionals

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Introduction

Iron is a mineral that is naturally present in many foods, added to some food products, and available as a dietary supplement. Iron is an essential component of hemoglobin, an erythrocyte (red blood cell) protein that transfers oxygen from the lungs to the tissues [1]. As a component of myoglobin, another protein that provides oxygen, iron supports muscle metabolism and healthy connective tissue [2]. Iron is also necessary for physical growth, neurological

development, cellular functioning, and the synthesis of some hormones [2,3].

Dietary iron has two main forms: heme and nonheme [1]. Plants and iron-fortified foods contain nonheme iron only, whereas meat, seafood, and poultry contain both heme and nonheme iron [2]. Heme iron, which forms when iron combines with protoporphyrin IX, contributes about 10% to 15% of total iron intakes in western populations [3-5].

Most of the 3 to 4 grams (g) of elemental iron that is present in adults is found in hemoglobin [2]. Much of the remaining iron is stored in the form of ferritin or hemosiderin (a degradation product of ferritin) in the liver, spleen, and bone marrow, or it is located in the myoglobin of muscle tissue [1,5].

Transferrin is the main protein in blood that binds to iron and transports it throughout the body. Humans typically lose only small amounts of iron in urine, feces, sweat, and shed skin cells.

Losses are greater in menstruating women because of blood loss.

Hepcidin, a circulating peptide hormone, is the key regulator of both iron absorption and the distribution of iron throughout the body, including in

plasma [1,2,6].

Assessing iron status



Hemoglobin and hematocrit are the most commonly used measures to screen patients for iron deficiency, although they are neither sensitive nor specific. Serum ferritin concentration, which is a measure of the body's iron stores, is also used, but it can be affected by inflammation. Often, health care providers will use multiple measurements to diagnose iron deficiency. They may also consider a patient's dietary and supplemental iron intakes and how those compare to intake recommendations.

The assessment of iron status depends almost entirely on hematological indicators [7]. However, these indicators are not sensitive or specific enough to adequately describe the full spectrum of iron status, and this can complicate the diagnosis of iron deficiency. A complementary approach is to consider how iron intakes from the diet and dietary supplements compare with recommended intakes.

Iron deficiency progresses from the

depletion of iron stores (mild iron deficiency), to iron-deficiency erythropoiesis (erythrocyte production), and finally to iron deficiency anemia (IDA) [8,9]. With iron-deficiency erythropoiesis (also known as marginal iron deficiency), iron stores are depleted and transferrin saturation declines, but hemoglobin levels are usually within the normal range. IDA is characterized by low hemoglobin concentrations and decreases in hematocrit (the proportion of red blood cells in blood by volume) and mean corpuscular volume (a measure of erythrocyte size) [2,10].

Serum ferritin concentration, a measure of the body's iron stores, is currently the most efficient and cost-effective test for diagnosing iron deficiency [11-13]. Because serum ferritin concentration decreases during the first stage of iron depletion, it can be used to identify low iron status before the onset of IDA [7,9,14]. A serum ferritin concentration that is lower than 30 micrograms/liter (mcg/L) suggests iron deficiency, and a value lower than 10 mcg/L suggests IDA [15]. However, it should be noted that inflammation (e.g., in those with infectious diseases) can elevate serum ferritin concentrations [16].

Hemoglobin and hematocrit tests are the most commonly used measures to screen patients for iron deficiency, even though they are neither sensitive nor specific [5,7,17]. Often, hemoglobin concentrations are combined with serum ferritin measurements to identify IDA [7]. Hemoglobin concentrations lower than 11 g/deciliter (dL) in children under 10 years of age, or lower than 12 g/dL in individuals age 10 years or older, suggest IDA [8]. Normal hematocrit values are approximately 41% to 50% in males and 36% to 44% in females [18].

Recommended Intakes



The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for iron. These values range from 8 to 27 mg for adults and from 0.27 to 27 mg for infants, children, and adolescents, depending on age, sex, and life stage. People who follow vegetarian diets need more iron than those who include animal products in their diet due to the decreased bioavailability of nonheme

iron from plant-based foods.

Intake recommendations for iron and other nutrients are provided in the Dietary Reference Intakes (DRIs) developed by the Food and Nutrition Board (FNB) at the National Academies of Sciences, Engineering, and Medicine [5]. DRI is the general term for a set of reference values used for planning and assessing nutrient intakes of healthy people. These values include the following:

- Recommended Dietary Allowance (RDA): Average daily level of intake sufficient to meet the nutrient requirements of nearly all (97%–98%) healthy individuals; often used to plan nutritionally adequate diets for individuals
- Adequate Intake (AI): Intake at this level is assumed to ensure nutritional adequacy; established when evidence is insufficient to develop an RDA
- Estimated Average Requirement (EAR): Average daily level of intake estimated to meet the requirements of 50% of healthy individuals; usually used to assess

the nutrient intakes of groups of people and to plan nutritionally adequate diets for them; can also be used to assess the nutrient intakes of individuals

- Tolerable Upper Intake Level (UL): Maximum daily intake unlikely to cause adverse health effects

Table 1 lists the current iron RDAs. The requirement for iron is 1.8 times higher for people who follow vegetarian diets than those for people who include animal products in their diet, such as meat. This is because heme iron from meat is more bioavailable than nonheme iron from plant-based foods. In addition, animal products such as meat, poultry, and seafood increase the absorption of nonheme iron [5].

For infants from birth to 6 months, the FNB established an AI for iron that is equivalent to the mean intake of iron in healthy, breastfed infants.

Table 1: Recommended Dietary Allowances (RDAs) for Iron in Milligrams (mg) [5]

Age	Male	Female	Pregnancy	Lactation
0–6 months	0.27 mg*	0.27 mg*		
7–12 months	11 mg	11 mg		

1–3 years	7 mg	7 mg		
4–8 years	10 mg	10 mg		
9–13 years	8 mg	8 mg		
14–18 years	11 mg	15 mg	27 mg	10 mg
19–50 years	8 mg	18 mg	27 mg	9 mg
51+ years	8 mg	8 mg		

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* Adequate Intake (AI)

Sources of Iron

Food



Lean meat and seafood are the richest dietary sources of heme iron, while nuts, beans, and vegetables contain nonheme iron. Wheat and other flours are often fortified with iron, making bread, cereal, and other grain products good dietary sources of nonheme iron as well. Heme iron has better bioavailability than nonheme iron; in addition, consuming other dietary components with nonheme iron, such

as ascorbic acid and phytate, can significantly affect the bioavailability of nonheme iron.

The richest sources of heme iron in the diet include lean meat and seafood [19]. Dietary sources of nonheme iron include nuts, beans, vegetables, and fortified grain products. In the United States, about half of dietary iron comes from bread, cereal, and other grain products [2,3,5]. Breast milk contains highly bioavailable iron, but the amounts found in breast milk are not sufficient to meet the needs of infants older than 4 to 6 months [2,20].

In the United States, Canada, and many other countries, wheat and other flours are fortified with iron [21,22]. Infant formulas are fortified with 12 milligrams (mg) iron per liter [20].

Heme iron has higher bioavailability than nonheme iron, and other dietary components have less effect on the bioavailability of heme than nonheme iron [3,4]. The bioavailability of iron is approximately 14% to 18% from mixed diets that include substantial amounts of meat, seafood, and vitamin C (ascorbic acid, which enhances the

bioavailability of nonheme iron) and 5% to 12% from vegetarian diets [2,4]. In addition to ascorbic acid, meat, poultry, and seafood can enhance nonheme iron absorption, whereas phytate (which is present in grains and beans) and certain polyphenols in some nonanimal foods (e.g., cereals, legumes) have the opposite effect [4]. Unlike other inhibitors of iron absorption, calcium might reduce the bioavailability of both nonheme and heme iron. However, the effects of enhancers and inhibitors of iron absorption are attenuated by a typical western diet, so they have little effect on most people's iron status.

Several food sources of iron are listed in Table 2. Some plant-based foods that are good sources of iron, such as spinach, have low iron bioavailability because they contain iron-absorption inhibitors, such as polyphenols [23,24].

Table 2: Iron Content of Selected Foods [25]

Food	Milligrams (mg)		Percent DV*
	per serving		
Breakfast cereals, fortified with 100% of the DV for iron, 1 serving	18		100

Oysters, eastern, cooked with moist heat, 3 ounces	8	44
White beans, canned, 1 cup	8	44
Beef liver, pan fried, 3 ounces	5	28
Lentils, boiled and drained, ½ cup	3	17
Spinach, boiled and drained, ½ cup	3	17
Tofu, firm, ½ cup	3	17
Chocolate, dark, 45%–69% cacao solids, 1 ounce	2	11
Kidney beans, canned, ½ cup	2	11
Sardines, Atlantic, canned in oil, drained solids with bone, 3 ounces	2	11
Chickpeas, boiled and drained, ½ cup	2	11
Tomatoes, canned, stewed, ½ cup	2	11
Beef, braised bottom round, with fat trimmed, 3 ounces	2	11
Potato, baked, flesh and skin, 1	2	11

medium potato		
Cashew nuts, oil roasted, 1 ounce (18 nuts)	2	11
Green peas, boiled, ½ cup	1	6
Chicken, roasted, meat and skin, 3 ounces	1	6
Rice, white, long grain, enriched, parboiled, drained, ½ cup	1	6
Bread, whole wheat, 1 slice	1	6
Bread, white, 1 slice	1	6
Raisins, seedless, ¼ cup	1	6
Spaghetti, whole wheat, cooked, 1 cup	1	6
Tuna, light, canned in water, 3 ounces	1	6
Turkey, roasted, breast meat and skin, 3 ounces	1	6
Nuts, pistachio, dry roasted, 1 ounce (49 nuts)	1	6
Broccoli, boiled	1	6

and drained, ½ cup

Egg, hard boiled, 1 large	1	6
Rice, brown, long or medium grain, cooked, 1 cup	1	6
Cheese, cheddar, 1.5 ounces	0	0
Cantaloupe, diced, ½ cup	0	0
Mushrooms, white, sliced and stir-fried, ½ cup	0	0
Cheese, cottage, 2% milk fat, ½ cup	0	0
Milk, 1 cup	0	0

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* DV = Daily Value. The U.S. Food and Drug Administration (FDA) developed DVs to help consumers compare the nutrient contents of foods and dietary supplements within the context of a total diet. The DV for iron is 18 mg for adults and children age 4 years and older [26]. FDA requires food labels to list iron content. Foods providing 20% or more of the DV are considered to be high sources of a nutrient, but foods providing lower percentages of the DV also contribute to a healthful diet.

The U.S. Department of Agriculture's (USDA's) FoodData Central (<https://fdc.nal.usda.gov/>) [25] lists the nutrient content of many foods and provides a comprehensive list of foods containing iron arranged by nutrient content and by food name.

Dietary supplements



Ferrous and ferric iron salts are the most common forms of iron found in dietary supplements, although other forms are also used. The various forms of iron contain different amounts of elemental iron, and certain forms may be more likely to cause gastrointestinal side effects at high doses. In addition, experts recommend taking calcium and iron supplements at different times to avoid potential interference with the absorption of iron.

Iron is available in many dietary supplements. Multivitamin/mineral supplements with iron, especially those designed for women, typically provide 18 mg iron, which is 100% of the Daily Value (DV). Multivitamin/mineral supplements for men or seniors

frequently contain less or no iron. Iron-only supplements usually deliver more than the DV, with many providing 65 mg iron (360% of the DV).

Frequently used forms of iron in supplements include ferrous and ferric iron salts, such as ferrous sulfate, ferrous gluconate, ferric citrate, and ferric sulfate [3,27]. Because of its higher solubility, ferrous iron in dietary supplements is more bioavailable than ferric iron [3]. High doses of supplemental iron (45 mg/day or more) may cause gastrointestinal side effects, such as nausea and constipation [5]. Other forms of supplemental iron, such as heme iron polypeptides, carbonyl iron, iron amino-acid chelates, and polysaccharide-iron complexes, might have fewer gastrointestinal side effects than ferrous or ferric salts [27].

The different forms of iron in supplements contain varying amounts of elemental iron. For example, ferrous fumarate is 33% elemental iron by weight, whereas ferrous sulfate is 20% and ferrous gluconate is 12% elemental iron [27]. Fortunately, elemental iron is listed in the Supplement Facts panel, so consumers do not need to calculate the amount of iron supplied by various

forms of iron supplements.

Approximately 14% to 18% of Americans use a supplement that contains iron [28,29]. The proportion of people who use supplements that contain iron varies by age and gender, ranging from 6% of children age 12 to 19 years to 60% of women who are lactating and 72% of pregnant women [28,30].

Calcium might interfere with the absorption of iron, although this effect has not been definitively established [4,31]. For this reason, experts suggest that people take individual calcium and iron supplements at different times of the day [32].

Iron Intakes and Status



While most people in the United States obtain adequate amounts of iron from their diets, certain factors, such as age, sex, race, and socioeconomic status, can put people at a higher risk of iron deficiency. At the other end of the spectrum, some people, including individuals with hereditary hemochromatosis, are at risk of

obtaining excess iron.

People in the United States usually obtain adequate amounts of iron from their diets, but infants, young children, teenage girls, pregnant women, and premenopausal women are at risk of obtaining insufficient amounts [28,33-35]. The average daily iron intake from foods is 11.5 to 13.7 mg/day in children age 2 to 11 years, 15.1 mg/day in children and teens age 12 to 19 years, and 16.3 to 18.2 mg/day in men and 12.6 to 13.5 mg/day in women older than 19 [28]. The average daily iron intake from foods and supplements is 13.7 to 15.1 mg/day in children age 2 to 11 years, 16.3 mg/day in children and teens age 12 to 19 years, and 19.3 to 20.5 mg/day in men and 17.0 to 18.9 mg/day in women older than 19. The median dietary iron intake in pregnant women is 14.7 mg/day [5].

Rates of iron deficiency vary by race and other sociodemographic factors. Six percent of White and Black toddlers age 1 to 3 years in the United States are iron deficient (defined as at least two abnormal results for the child's age and gender on transferrin saturation tests, free erythrocyte protoporphyrin tests,

and/or serum ferritin tests), compared with 12% of Hispanic toddlers [36]. Deficiency (including IDA) is more common among children and adolescents in food-insecure households than in food-secure households [36,37]. Among pregnant women, iron deficiency (based on depleted iron stores) is more common in Mexican American (23.6%) and non-Hispanic Black women (29.6%) than in non-Hispanic White women (13.9%) [38].

Some groups are at risk of obtaining excess iron. Individuals with hereditary hemochromatosis, which predisposes them to absorb excessive amounts of dietary iron, have an increased risk of iron overload [39]. One study suggests that elderly people are more likely to have chronic positive iron balance and elevated total body iron than iron deficiency. Among 1,106 elderly White adults age 67 to 96 years in the Framingham Heart Study, 13% had high iron stores (serum ferritin levels higher than 300 mcg/L in men and 200 mcg/L in women), of which only 1% was due to chronic disease [40]. The authors did not assess genotypes, so they could not determine whether these results were due to hemochromatosis [40].

Iron Deficiency



There are several stages of iron depletion and deficiency; iron deficiency anemia occurs when the body's iron stores are exhausted, and hematocrit and levels of hemoglobin decline. People with iron deficiency anemia may experience gastrointestinal disturbances, weakness, fatigue, and difficulty concentrating. The condition may also impair cognitive function, immune function, and body temperature regulation. People with iron deficiency often have other nutrient deficiencies, which can cause other types of anemia or affect the severity of anemia.

Iron deficiency is not uncommon in the United States, especially among young children, women of reproductive age, and pregnant women. Because iron deficiency is associated with poor diet, malabsorptive disorders, and blood loss, people with iron deficiency usually have other nutrient deficiencies [2]. The World Health Organization (WHO) estimates that approximately half of the 1.62 billion cases of anemia worldwide are due to iron deficiency [41]. In

developing countries, iron deficiency is often caused by enteropathies and blood loss that are associated with gastrointestinal parasites [2].

Iron depletion and deficiency progresses through several stages [8-10]:

1. Mild deficiency or storage iron depletion: Serum ferritin concentrations and levels of iron in bone marrow decrease.
2. Marginal deficiency, mild functional deficiency, or iron-deficient erythropoiesis (erythrocyte production): Iron stores are depleted and iron supply to erythropoietic cells and transferrin saturation declines, but hemoglobin levels are usually within the normal range.
3. IDA: Iron stores are exhausted; hematocrit and levels of hemoglobin decline; and the resulting microcytic, hypochromic anemia is characterized by small red blood cells with low hemoglobin concentrations.

In 2002, the WHO characterized IDA as one of the 10 leading risk factors for

disease around the world [42]. Although iron deficiency is the most common cause of anemia, deficiencies of other micronutrients (e.g., folate, vitamin B12) and other factors (e.g., chronic infection and inflammation) can cause different forms of anemia or contribute to their severity.

The functional deficits that are associated with IDA include gastrointestinal disturbances; weakness; fatigue; difficulty concentrating; and impaired cognitive function, immune function, exercise or work performance, and body temperature regulation [15,43]. In infants and children, IDA can result in psychomotor and cognitive abnormalities that, without treatment, can lead to learning difficulties [2,43]. Some evidence indicates that the effects of deficiencies early in life persist through adulthood [2]. Because iron deficiency is often accompanied by deficiencies in other nutrients, the signs and symptoms of iron deficiency can be difficult to isolate [2].

Groups at Risk of Iron Inadequacy



Certain groups of people are more likely than others to have inadequate iron intakes, including pregnant women, infants and young children, women with heavy menstrual bleeding, and frequent blood donors. People with certain conditions, such as cancer, gastrointestinal disorders, and heart failure, may also have inadequate iron intakes.

The following groups are among those most likely to have inadequate intakes of iron.

Pregnant women

During pregnancy, plasma volume and red cell mass expand as maternal red blood cell production increases to meet the needs of the fetus and placenta [2]. As a result of this expansion, the amount of iron that women need increases during pregnancy. Iron deficiency during pregnancy increases the risk of maternal and infant mortality, premature birth, and low birthweight [44].

Infants and young children

Infants—especially those born preterm or with low birthweight or whose

mothers have iron deficiency—are at risk of iron deficiency because of their high iron requirements due to their rapid growth [34,45]. Full-term infants usually have sufficient iron stores and need little if any iron from external sources until they are 4 to 6 months old [2]. However, full-term infants have a risk of becoming iron deficient at 6 to 9 months unless they obtain adequate amounts of solid foods that are rich in bioavailable iron or iron-fortified formula.

Women with heavy menstrual bleeding

Women of reproductive age who have menorrhagia, or abnormally heavy bleeding during menstruation, have an increased risk of iron deficiency. At least 10% of menstruating women are believed to have menorrhagia, but the percentage varies widely depending on the diagnostic criteria used [46-48].

Women with menorrhagia lose significantly more iron per menstrual cycle on average than women with normal menstrual bleeding [49]. Limited evidence suggests that menorrhagia might be responsible for about 33% to 41% of cases of IDA in women of reproductive age [50,51].

Frequent blood donors

Frequent blood donors have an increased risk of iron deficiency [5]. In the United States, adults may donate blood as often as every 8 weeks, which can deplete iron stores. About 25% to 35% of regular blood donors develop iron deficiency [52]. In a study of 2,425 blood donors, men who had given at least three whole-blood donations and women who had given at least two whole-blood donations in the previous year were more than five times as likely to have depleted iron stores as first-time donors [53]. A clinical trial of iron supplementation found that of 215 adults who had donated a unit of blood within the past 3 to 8 days, those randomized to take an iron supplement (37.5 mg/day elemental iron from ferrous gluconate) for 24 weeks recovered their lost hemoglobin and iron in less than half the time of those not given the supplement [52]. At 24 weeks, two-thirds of the donors who did not receive iron supplementation had not recovered the iron had they lost.

People with cancer

Up to 60% of patients with colon cancer have iron deficiency at diagnosis,

probably due to chronic blood loss [54]. The prevalence of iron deficiency in patients with other types of cancer ranges from 29% to 46%. The main causes of iron deficiency in people with cancer are anemia of chronic disease (discussed in the **Iron and Health** section below) and chemotherapy-induced anemia. However, chronic blood loss and deficiencies of other nutrients (due, for example, to cancer-induced anorexia) can exacerbate iron deficiency in this population.

People who have gastrointestinal disorders or have had gastrointestinal surgery

People with certain gastrointestinal disorders (e.g., celiac disease, ulcerative colitis, Crohn's disease) or who have undergone certain gastrointestinal surgical procedures (e.g., gastrectomy, intestinal resection) have an increased risk of iron deficiency because their disorder or surgery requires dietary restrictions or results in iron malabsorption or blood loss in the gastrointestinal tract [55-57]. The combination of low iron intake and high iron loss can lead to a negative iron balance; reduced production of hemoglobin; or microcytic, hypochromic

anemia [58].

People with heart failure

Approximately 60% of patients with chronic heart failure have iron deficiency and 17% have IDA, which might be associated with a higher risk of death in this population [59,60].

Potential causes of iron deficiency in people with heart failure include poor nutrition; malabsorption; defective mobilization of iron stores; cardiac cachexia; and the use of aspirin and oral anticoagulants, which might result in the loss of some blood in the gastrointestinal tract [61].

Iron and Health

This section focuses on the role of iron in IDA in pregnant women, infants, and toddlers as well as in anemia of chronic disease.

Iron deficiency anemia in pregnant women



Iron supplementation has been shown to reduce the risk of iron deficiency anemia in pregnant women and the risk of certain adverse effects in their

infants. Accordingly, some professional societies and public health organizations recommend anemia screening and iron supplementation for pregnant women, but not all groups agree that the evidence is sufficient to recommend routine screening and supplementation for asymptomatic pregnant women.

Insufficient iron intakes during pregnancy increase a woman's risk of IDA [62-65]. Low intakes also increase her infant's risk of low birthweight, premature birth, low iron stores, and impaired cognitive and behavioral development.

An analysis of 1999–2006 data from the National Health and Nutrition Examination Survey (NHANES) found that 18% of pregnant women in the United States had iron deficiency [38]. Rates of deficiency were 6.9% among women in the first trimester, 14.3% in the second trimester, and 29.7% in the third trimester.

Randomized controlled trials have shown that iron supplementation can prevent IDA in pregnant women and related adverse effects in their infants

[66,67]. A Cochrane Review showed that daily supplementation with 9 to 90 mg iron reduced the risk of anemia in pregnant women at term by 70% and of iron deficiency at term by 57% [64]. In the same review, use of daily iron supplements was associated with an 8.4% risk of having a low-birthweight newborn compared to 10.2% with no supplementation. In addition, the mean birthweight was 31 g higher for infants whose mothers took daily iron supplements during pregnancy than for the infants of mothers who did not take iron.

The guidelines from professional societies and public health organizations vary on iron supplementation during pregnancy:

- The American College of Obstetricians and Gynecologists (ACOG) recommends low-dose iron supplementation (dose not specified) starting in the first trimester to decrease the prevalence of maternal anemia at delivery [68]. It also recommends screening all pregnant women for anemia in the first trimester and again at 24 to 28 weeks gestation and evaluating those who

have anemia (defined as hematocrit levels less than 33% in the first and third trimesters and less than 32% in the second trimester) to determine the cause. ACOG states that IDA during pregnancy is associated with increased risks of preterm birth, low birth weight, and perinatal mortality and should be treated with iron supplementation in addition to prenatal vitamins.

- In contrast, the U.S. Preventive Services Task Force (USPSTF) has concluded that the current evidence is insufficient to assess the balance of benefits and harms of both screening for iron deficiency and IDA in pregnant women and routinely supplementing them with iron to prevent adverse maternal and infant health outcomes [69]. They note, however, that their recommendation applies to asymptomatic pregnant women. It does not apply to pregnant women who are severely malnourished, have symptoms of iron deficiency or IDA, or have conditions such as sickle cell disease or nutritional deficiencies that may increase iron

needs.

- The FNB notes that because the median intake of dietary iron by pregnant women is well below the EAR, pregnant women need iron supplementation [5].
- The *Dietary Guidelines for Americans* advises that women who are pregnant take an iron supplement when recommended by an obstetrician or other health care provider [19]. It adds that low intakes of iron are a public health concern for pregnant women.

Iron deficiency anemia in infants and toddlers



In infants, iron deficiency anemia can cause adverse cognitive and psychological effects that are potentially irreversible. Studies indicate that iron supplementation or fortification of foods with iron can reduce the risk of iron deficiency in infants and young children, but guidelines from professional societies and public health organizations vary for this population.

Approximately 18% of infants age 6 to 12 months in the United States have inadequate iron intakes, and 8% of toddlers have iron deficiency [36,70]. The prevalence of IDA in U.S. toddlers age 12 to 35 months ranges from 0.9% to 4.4% depending on race or ethnicity and socioeconomic status [20]. Full-term infants typically have adequate iron stores for approximately the first 4 to 6 months, but the risk of iron deficiency in low-birthweight and preterm infants begins at birth because of their low iron stores.

IDA in infancy can lead to adverse cognitive and psychological effects, including delayed attention and social withdrawal; some of these effects might be irreversible [2,20]. In addition, IDA is associated with higher lead concentrations in the blood (although the cause of this is not fully understood), which can increase the risk of neurotoxicity [20].

A Cochrane Review of 26 studies in 2,726 preterm and low-birthweight infants found that enteral iron supplementation (at least 1 mg/kg/day) reduces the risk of iron deficiency, but the long-term effects of supplementation on

neurodevelopmental outcomes and growth is not clear [71]. Another Cochrane Review of eight trials in 3,748 children younger than 2 in low-income countries showed that home fortification of semi-solid foods with micronutrient powders containing 12.5 mg to 30 mg elemental iron as ferrous fumarate and 4 to 14 other micronutrients for 2 to 12 months reduced rates of anemia by 31% and of iron deficiency by 51% compared with no intervention or placebo but had no effect on any growth measurements [72].

The guidelines from professional societies and public health organizations vary on iron supplementation for infants and young children:

- The Centers for Disease Control and Prevention (CDC) recommends that infants less than 12 months of age who are not exclusively or primarily breastfed drink iron-fortified infant formula [17]. Breastfed infants who were born preterm or with a low birthweight should receive 2 to 4 mg/kg/day of iron drops (to a maximum of 15 mg/day) from ages 1 to 12 months.

Breastfed infants who receive insufficient iron (less than 1 mg/kg/day) from supplementary foods by age 6 months should receive 1 mg/kg/day of iron drops. The CDC also recommends that infants and preschool children at high risk for IDA (e.g., children from low-income families, migrant children) be screened between age 9 to 12 months, 6 months later, and annually from age 2 to 5 years. Treatment for IDA begins with 3 mg/kg/day of iron drops given between meals.

- The American Academy of Pediatrics recommends 1 mg/kg daily iron supplementation for exclusively or primarily breastfed full-term infants from age 4 months until the infants begin eating iron-containing complementary foods, such as iron-fortified cereals [20]. Standard infant formulas that contain 10 to 12 mg/L of iron can meet the iron needs of infants for the first year of life. The Academy recommends 2 mg/kg/day iron supplementation for preterm infants age 1 to 12 months who are fed breast milk.

- The WHO recommends universal supplementation with 2 mg/kg/day of iron in children age 6 to 23 months whose diet does not include foods fortified with iron or who live in regions (e.g., developing countries) where anemia prevalence is higher than 40% [44].
- In a recommendation statement issued in 2015, the USPSTF concluded that the available evidence is insufficient to recommend for or against routine screening for IDA in children age 6 to 24 months who live in the United States and who are asymptomatic for IDA [73]. It added that this recommendation does not apply to severely malnourished children or children who were born prematurely or with low birthweight. Earlier, in 2006, the USPSTF stated that although it found insufficient evidence to recommend routine iron supplementation in asymptomatic infants at average risk of IDA, it did recommend routine iron supplements for children age 6 to 12 months who are at increased risk of IDA (e.g., those who were premature or low birthweight) [74].

The USPSTF's 2015 statement notes that its current recommendation is limited to screening because the widespread use of iron-fortified foods in the United States (including infant formulas and cereals) would likely limit the impact of iron supplementation prescribed by physicians [73].

Some studies have suggested that iron supplementation in young children living in areas where malaria is endemic could increase their risk of malaria [75,76]. However, a Cochrane Review of 33 trials in 13,114 children showed that intermittent supplementation does not appear to have this effect [77]. The WHO therefore recommends 6-month supplementation cycles as follows: children age 24 to 59 months should receive 25 mg iron and those age 5 to 12 years should receive 45 mg every week for 3 months, followed by 3 months of no supplementation [75]. The WHO recommends providing these supplements in malaria-endemic areas in conjunction with measures to prevent, diagnose, and treat malaria.

Anemia of chronic disease



Anemia of chronic disease is caused by certain inflammatory, infectious, and neoplastic diseases that disrupt iron homeostasis and limit the amount of iron that is available for erythropoiesis. Treating anemia of chronic disease is usually a matter of treating the underlying disease, but in some cases, patients may receive iron supplementation or erythropoiesis-stimulating agents. However, using iron supplementation to treat this condition remains controversial due to the risk of infection and cardiovascular events.

Certain inflammatory, infectious, and neoplastic diseases (e.g., rheumatoid arthritis, inflammatory bowel disease, hematologic malignancies) can cause anemia of chronic disease, also known as anemia of inflammation [2,78].

Anemia of chronic disease is the second most common type of anemia after IDA [79]. In people with anemia of chronic disease, inflammatory cytokines upregulate the hormone hepcidin. As a result, iron homeostasis is disrupted and iron is diverted from the circulation to storage sites, limiting the amount of iron available for erythropoiesis.

Anemia of chronic disease is usually mild to moderate (hemoglobin levels 8–9.5 g/dL) and is associated with low counts of erythrocytes and decreased erythropoiesis [78]. The condition can be difficult to diagnose; although low serum ferritin levels indicate iron deficiency, these levels tend to be higher in patients with infection or inflammation [80].

The clinical implications of iron deficiency in people with chronic diseases are not clear. Even mild anemia of chronic disease is associated with an increased risk of hospitalization and mortality in older people [81]. Two prospective observational studies found that iron deficiency in patients with objectively measured heart failure was associated with an increased risk of heart transplantation and death, and this association was independent of other well-established prognostic factors for poor outcomes, including anemia [82,83]. However, an analysis of NHANES data from 574 adults with self-reported heart failure found no association between iron deficiency and all-cause or cardiovascular mortality [60].

The main therapy for anemia of chronic

disease is treatment of the underlying disease [79]. But when such treatment is not possible, iron supplementation and/or erythropoiesis-stimulating agents (ESAs) are sometimes used. The use of iron supplements—whether oral, intravenous, or parenteral—in this setting is controversial because they might increase the risk of infection and cardiovascular events and could cause tissue damage [79].

Only a few small studies have evaluated the benefits of oral iron supplementation alone or in combination with ESAs to treat anemia of chronic disease. For example, a prospective observational study in 132 patients with anemia and chronic kidney disease who were not on dialysis or ESAs found that oral supplements (130 mg/day elemental iron from ferrous sulfate twice daily) for 1 year resulted in a decline in hemoglobin of only 0.13 g/dL compared with a decline of 0.46 g/dL in the placebo group [76,84]. In a randomized trial of 100 patients with cancer-related anemia, taking oral iron supplements (equivalent to 200 mg/day elemental iron, form of iron not specified) once per week with an ESA resulted in a mean increase of 2.4 g/dL hemoglobin after 24 weeks compared

with taking only oral supplements [85]. Iron administered parentally increases hemoglobin levels to a greater extent and is associated with fewer side effects than oral iron supplementation in patients with anemia of chronic disease [86].

Health Risks from Excessive Iron



The risk of iron overload from dietary sources of iron is low among adults who have normal intestinal function, but high doses of iron supplements can cause a range of gastrointestinal effects. With extremely high doses, these effects can be severe, including corrosive necrosis of the intestine, multisystem organ failure, and even death. The Tolerable Upper Intake Level for iron is 45 mg for adults, and it ranges from 40 mg to 45 mg for infants, children, and adolescents, depending on age.

Adults with normal intestinal function have very little risk of iron overload from dietary sources of iron [2]. However, supplements containing 25 mg iron or more can reduce zinc absorption and

plasma zinc concentrations [3,87,88]. High-dose iron supplements can also cause gastrointestinal effects, including gastric upset, constipation, nausea, abdominal pain, vomiting, and diarrhea [5,89]. Taking iron supplements with food can help minimize these adverse effects. Case reports, some of which involved doses of 130 mg iron, suggest that some people develop even more serious gastrointestinal effects, including gastritis and gastric lesions (along with iron deposits in the gastric mucosa in some cases) [90-93].

Acute intakes of more than 20 mg/kg iron (about 1,365 mg iron for a person weighing 150 pounds [lb]) from supplements or medicines can lead to corrosive necrosis of the intestine, which might lead to fluid and blood loss, shock, tissue damage, and organ failure, especially if food is not taken at the same time as the iron [89]. In severe cases (e.g., one-time ingestions of 60 mg/kg, or about 4,090 mg iron for a 150-lb person), overdoses of iron can lead to multisystem organ failure, coma, convulsions, and even death [27,94].

Between 1983 and 2000, at least 43 U.S. children died from ingesting supplements containing high doses of

iron (36–443 mg iron/kg body weight) [27]. Accidental ingestion of iron supplements caused about a third of poisoning deaths among children reported in the United States between 1983 and 1991.

In 1997, FDA began requiring oral supplements containing more than 30 mg elemental iron per dose to be sold in single-dose packaging with strong warning labels. At the same time, many manufacturers voluntarily replaced the sugar coating on iron tablets with film coatings. Between 1998 and 2002, only one child death due to ingesting an iron-containing tablet was reported [27]. As a result of a court decision, FDA removed its single-dose packaging requirement for iron supplements in 2003 [95]. FDA currently requires that iron-containing dietary supplements sold in solid form (e.g., tablets or capsules but not powders) carry the following label statement: “WARNING: Accidental overdose of iron-containing products is a leading cause of fatal poisoning in children under 6. Keep this product out of reach of children. In case of accidental overdose, call a doctor or poison control center immediately” [96]. In addition, since 1978, the Consumer Product Safety Commission has

required manufacturers to package dietary supplements that contain 250 mg or more elemental iron per container in child-resistant bottles or packaging to prevent accidental poisoning [97,98].

Hemochromatosis, a disease caused by a mutation in the hemochromatosis (HFE) gene, is associated with an excessive buildup of iron in the body [3,39,99]. About 1 in 10 Whites carry the most common HFE mutation (C282Y), but only 4.4 Whites per 1,000 are homozygous for the mutation and have hemochromatosis [100]. The condition is much less common in other ethnic groups. Without treatment by periodic chelation or phlebotomy, people with hereditary hemochromatosis typically develop signs of iron toxicity by their 30s [3]. These effects can include liver cirrhosis, hepatocellular carcinoma, heart disease, and impaired pancreatic function. The American Association for the Study of Liver Diseases recommends that treatment of hemochromatosis include the avoidance of iron and vitamin C supplements [39].

The FNB has established ULs for iron from food and supplements based on the amounts of iron that are associated

with gastrointestinal effects following supplemental intakes of iron salts (see Table 3). The ULs apply to healthy infants, children, and adults. Physicians sometimes prescribe intakes higher than the UL, such as when people with IDA need higher doses to replenish their iron stores [5].

Table 3: Tolerable Upper Intake Levels (ULs) for Iron [5]

Age	Male	Female	Pregnancy	Lactation
0–6 months	40 mg	40 mg		
7–12 months	40 mg	40 mg		
1–3 years	40 mg	40 mg		
4–8 years	40 mg	40 mg		
9–13 years	40 mg	40 mg		
14–18 years	45 mg	45 mg	45 mg	45 mg
19–50 years	45 mg	45 mg	45 mg	45 mg
51+ years	45 mg	45 mg		

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Interactions with Medications



Iron supplements may interact with medications, including levodopa and levothyroxine. In addition, proton pump inhibitors can potentially reduce iron absorption.

Iron can interact with certain medications, and some medications can have an adverse effect on iron levels. A few examples are provided below. Individuals taking these and other medications on a regular basis should discuss their iron status with their health care providers.

Levodopa

Some evidence indicates that in healthy people, iron supplements reduce the absorption of levodopa (found in Sinemet and Stalevo), used to treat Parkinson's disease and restless leg syndrome, possibly through chelation [101-103]. In the United States, the labels for levodopa warn that iron-containing dietary supplements might reduce the amount of levodopa available to the body and, thus, diminish

its clinical effectiveness [104,105].

Levothyroxine

Levothyroxine (Levothroid, Levoxyl, Synthroid, Tirosint, and Unithroid) is used to treat hypothyroidism, goiter, and thyroid cancer. The simultaneous ingestion of iron and levothyroxine can result in clinically significant reductions in levothyroxine efficacy in some patients [106]. The labels for some of these products [107,108] warn that iron supplements can reduce the absorption of levothyroxine tablets and advise against administering levothyroxine within 4 hours of iron supplements.

Proton pump inhibitors

Gastric acid plays an important role in the absorption of nonheme iron from the diet. Because proton pump inhibitors, such as lansoprazole (Prevacid) and omeprazole (Prilosec), reduce the acidity of stomach contents, they can reduce iron absorption [3].

Treatment with proton pump inhibitors for up to 10 years is not associated with iron depletion or anemia in people with normal iron stores [109] but patients with iron deficiency taking proton pump inhibitors can have suboptimal

responses to iron supplementation [110].

Iron and Healthful Diets



In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

For more information about building a healthy dietary pattern, refer to the *Dietary Guidelines for Americans* (<https://www.dietaryguidelines.gov>).

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